

VICTORIA SUN

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🐙 github.com/victoriasunshine portfolio : victoriasun.xyz

Education

University of California, Irvine

Bachelor of Science in Computer Science

Irvine Valley College

Associates with a focus in Natural Sciences

Sep. 2025 – June 2028

Irvine, CA

Aug. 2023 – May 2025

Irvine, CA

Projects

Legacy Robotics: Panoramic Rover Mapping System | C++, OpenCV, ROS2, ZED2 - [Github](#) April 2026-Present

- Own the panoramic image stitching pipeline for a legacy rover platform: cylindrical projection warping, SIFT feature detection, and RANSAC-based homography estimation to chain multi-frame image sequences into a single wide-field panorama.
- Designed and implemented a ROS2 (Humble) node that integrates the stitching pipeline into the rover's live camera feed, buffering frames from the ZED2 stereo camera and exposing an on-demand stitch service, replacing the original offline/batch-only workflow.
- Built a post-processing stage for seam and hole correction using connected-component analysis and inpainting to reconstruct clean panorama boundaries after multi-image blending.
- Hands-on experience using Git, setting up a team Docker, and working with an NVIDIA-Jetson

Personal Portfolio Site | victoriasun.xyz

January 2026-Present

- Built and deployed a full-stack personal site using Next.js and TypeScript, with Tailwind CSS for styling and Vercel for CI/CD and hosting.
- Integrated Sanity CMS via its API to drive dynamic content (projects, writing) without hardcoding page content, allowing the site to update independent of code deploys.

Foundation Fix | Python — Paper — [Github](#)

September 2025-December 2025

- Developed a computer vision-based color calibration system using OpenCV to extract and correct skin tone measurements under varying lighting.
- Implemented color space transformations (RGB → XYZ → CIE Lab) and reference-based lighting correction using a Macbeth ColorChecker.
- Authored a full engineering design report documenting requirements, iterations, algorithms, circuitry, and system tradeoffs.

Experience

UCI Theta Tau – Pi Delta Chapter

June 2026 – Present

Scribe, Executive Board

Irvine, CA

- Elected to the executive board as chapter Scribe; record and maintain meeting minutes and chapter records.
- Manage internal chapter communications, including Cabinet Board, and maintain the chapter's shared Google Calendar for events and officer meetings.

Bay Honors Symposium

November 2025 – Present

Researcher, Presenter, Author — Paper

Berkeley, CA

- Performed data analysis on a cosmic ray muon detector experiment (based on the MIT CosmicWatch design), calculating muon flux rates from Arduino-logged detection data; authored the full research paper detailing methodology, results, and theoretical basis, achieving an honorable mention in the Heslet Scholar Award.
- Led the presentation group in constructing the slides, drafting outlines for the presentation, and organizing practice speeches leading up to the conference. Achieved an honorable mention in the Heslet Scholar Award, an award for the best project in the conference, based on abstract and presentation. Honorable mention is given to five groups.

Irvine Valley College: Mathematics, Engineering, Science Achievement February 2025 – August 2025

Physics, Math, CS Tutor

Irvine, CA

- Proctored quizzes for classes of 30 and gave live feedback to students. Quizzes in this class were collaborative and I was permitted to give incremental pushes in the right direction to encourage independent thought.
- Held private office hours in which I helped students review lecture or textbook material, prepare for tests, and taught methods in finding and fixing bugs in code.

Skills

Languages/Frameworks: C/C++, Java, HTML, CSS, JavaScript, Python, TypeScript

Frameworks/Libraries: OpenCV, ROS2, Next.js, Tailwind CSS, Sanity CMS **Developer Tools:** VS Code, XCode, R Studio, JetBrains, Git, Github

Spoken Languages: English, Chinese (Mandarin)